

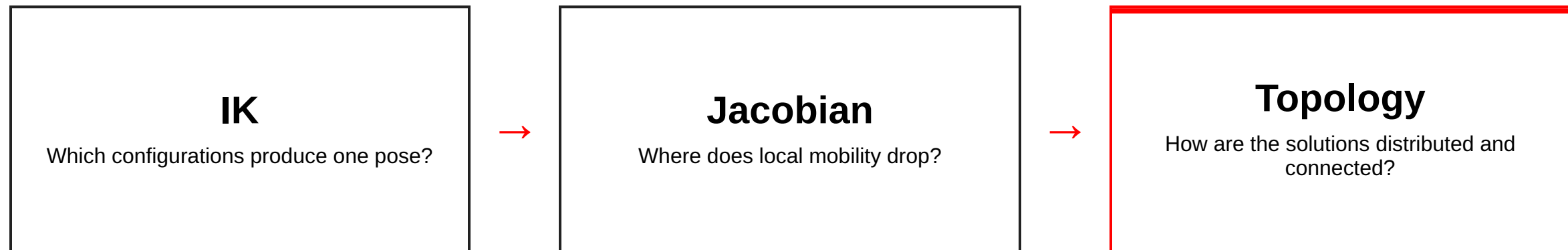
ENG-654 · LECTURE 06

Cuspidality

How inverse-kinematic solutions are connected across joint space

From singularity curves to IK-solution distribution, aspects, and cuspidality—with 3R, wrist-partitioned 6R, ABB IRB 4600, and FANUC examples.

Singularities organize the global distribution of IK solutions

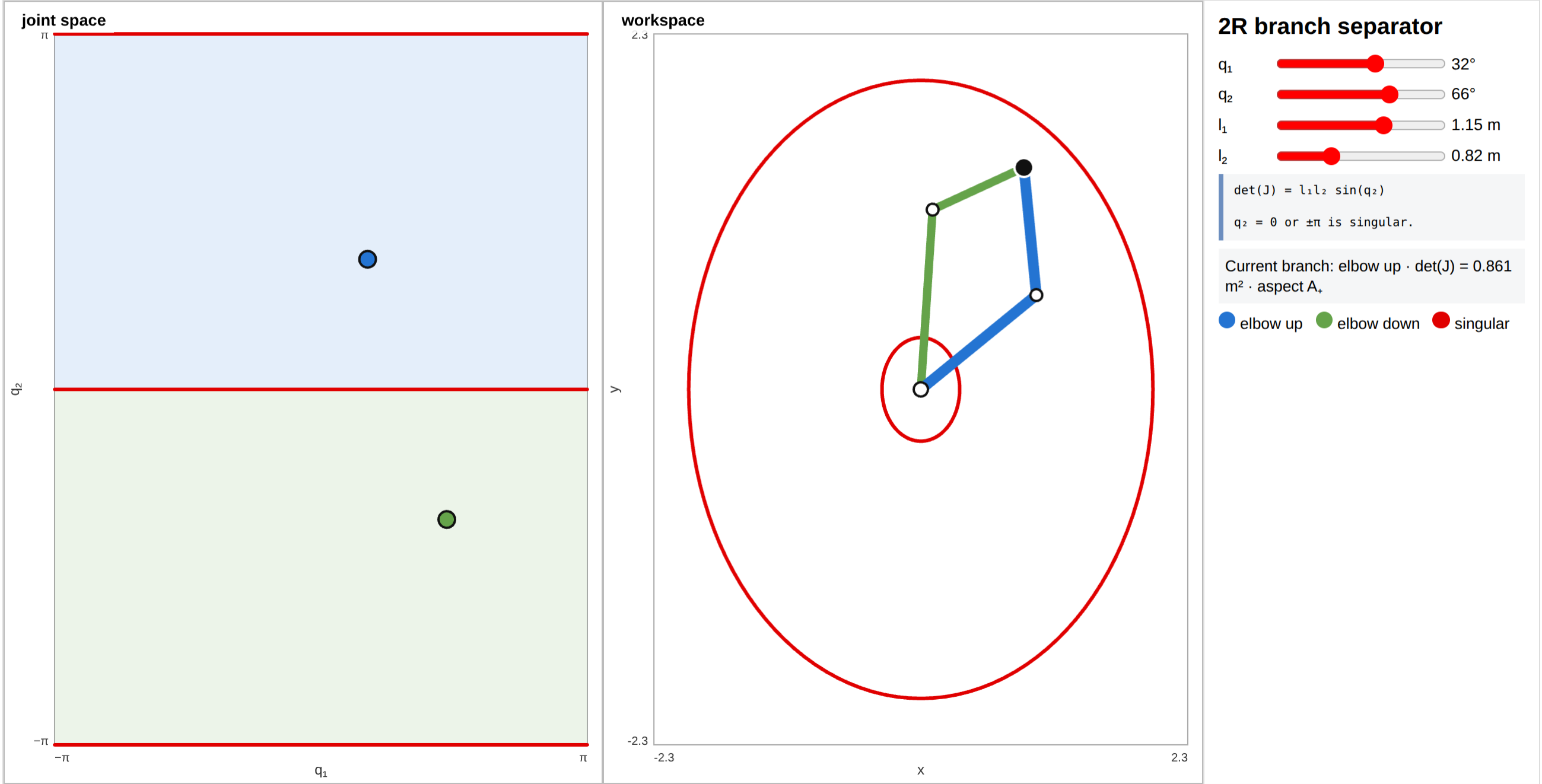


New question: can a robot move from one IK solution to another without crossing a singularity?

A 2R robot separates its two IK branches with singular lines

The determinant depends only on the relative elbow angle. Move the target or either joint and compare the two maps.

Explore: drag in either plot and observe that elbow-up and elbow-down meet only at a singular workspace boundary.



Aspects are maximal connected singularity-free regions

Remove the singular configurations from joint space.

$$\mathcal{S} = \{\mathbf{q} : \text{rank } J(\mathbf{q}) < r_{\max}\}$$

Each maximal connected region that remains is an **aspect**:

$$\mathcal{A}_i \subset \mathcal{C} \setminus \mathcal{S}$$

Here $r_{\max}$ is the maximum attainable rank. For a square, generically full-rank Jacobian, singularity means $\det J = 0$. For a rectangular Jacobian, check rank or singular values.

Inside one aspect

Every pair of configurations can be joined by at least one singularity-free joint path.

Across aspects

Every connecting joint path intersects the singularity set.

Feasible components

The sign of $\det J$ is only a diagnostic. Joint limits and collisions can further split one mathematical aspect into disconnected feasible regions.

Cuspidality means two IK solutions can share one aspect

A robot is **cuspidal** if two distinct inverse-kinematic solutions of the same pose can be connected by a path that contains no singular configuration.

$$\mathbf{q}_a \neq \mathbf{q}_b, \quad f(\mathbf{q}_a) = f(\mathbf{q}_b), \quad \exists \gamma : [0, 1] \rightarrow \mathcal{A}_i \text{ with } \gamma(0) = \mathbf{q}_a, \gamma(1) = \mathbf{q}_b.$$

The joint path is open, while its image under forward kinematics is a closed loop in task space.

The atlas determinant and cylindrical coordinates both eliminate q_1

Offset-atlas determinant

$$\det J_p = \frac{3}{4}(3c_3 + 4)[c_2(c_3 - 2s_3) - s_3]$$

For the idealized offset D–H preset used in the atlas, the expression contains q_2 and q_3 , but not q_1 . A plotted zero therefore represents a circle of singular configurations parameterized by q_1 .

Meridian reduction

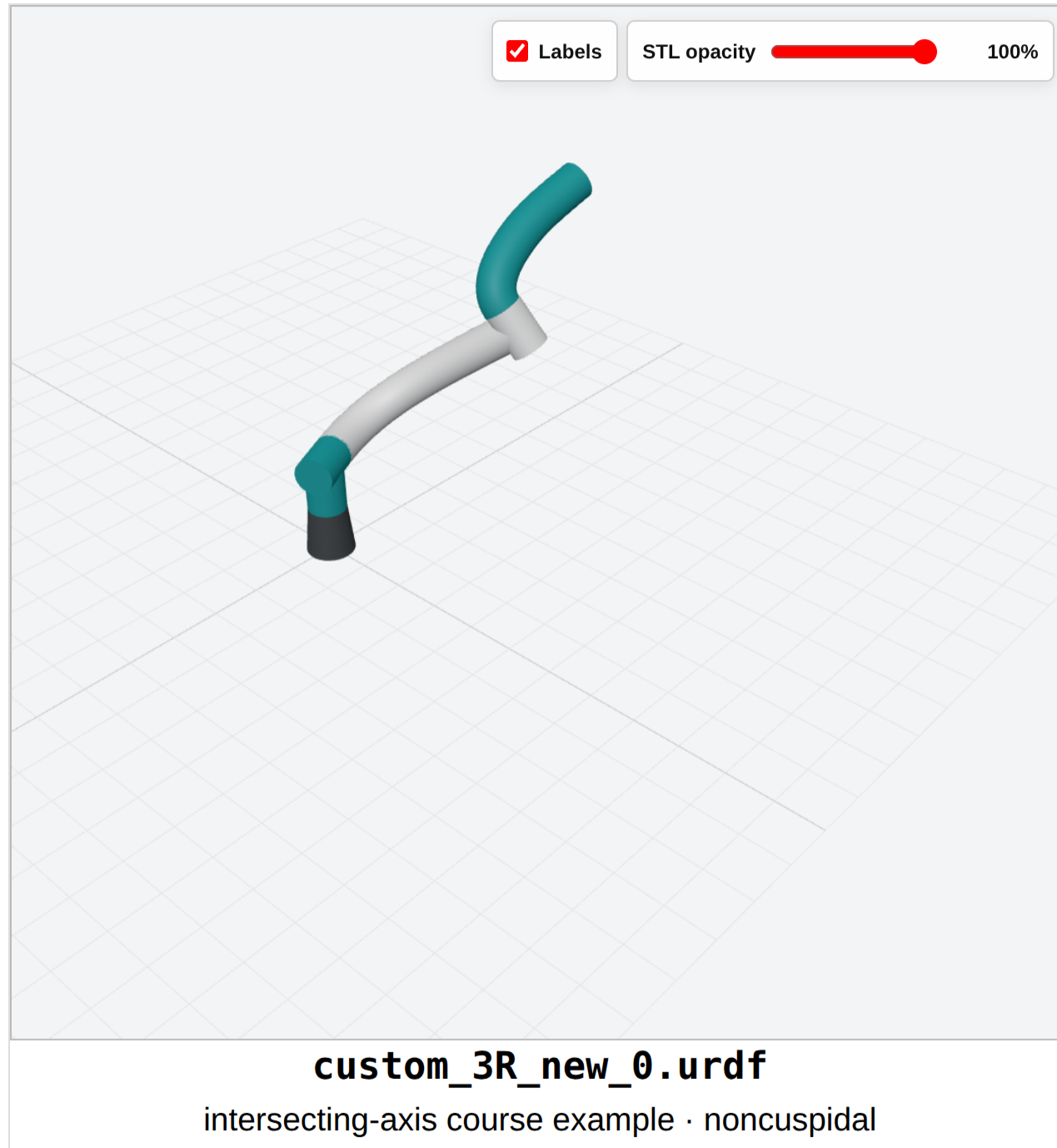
$$\begin{bmatrix} x \\ y \end{bmatrix} = R(q_1) \begin{bmatrix} u(q_2, q_3) \\ v(q_2, q_3) \end{bmatrix}$$

$$\rho^2 = x^2 + y^2 = u^2 + v^2, \quad z = w(q_2, q_3)$$

The rotation $R(q_1)$ disappears from ρ . We may set $y = 0$, use $x = \rho$, and study the (ρ, z) slice without losing singularity information.

For this axisymmetric teaching model, joint-space singular curves live in (q_2, q_3) ; their forward-kinematic images are critical curves in (ρ, z) .

Studying the examples from previous lectures

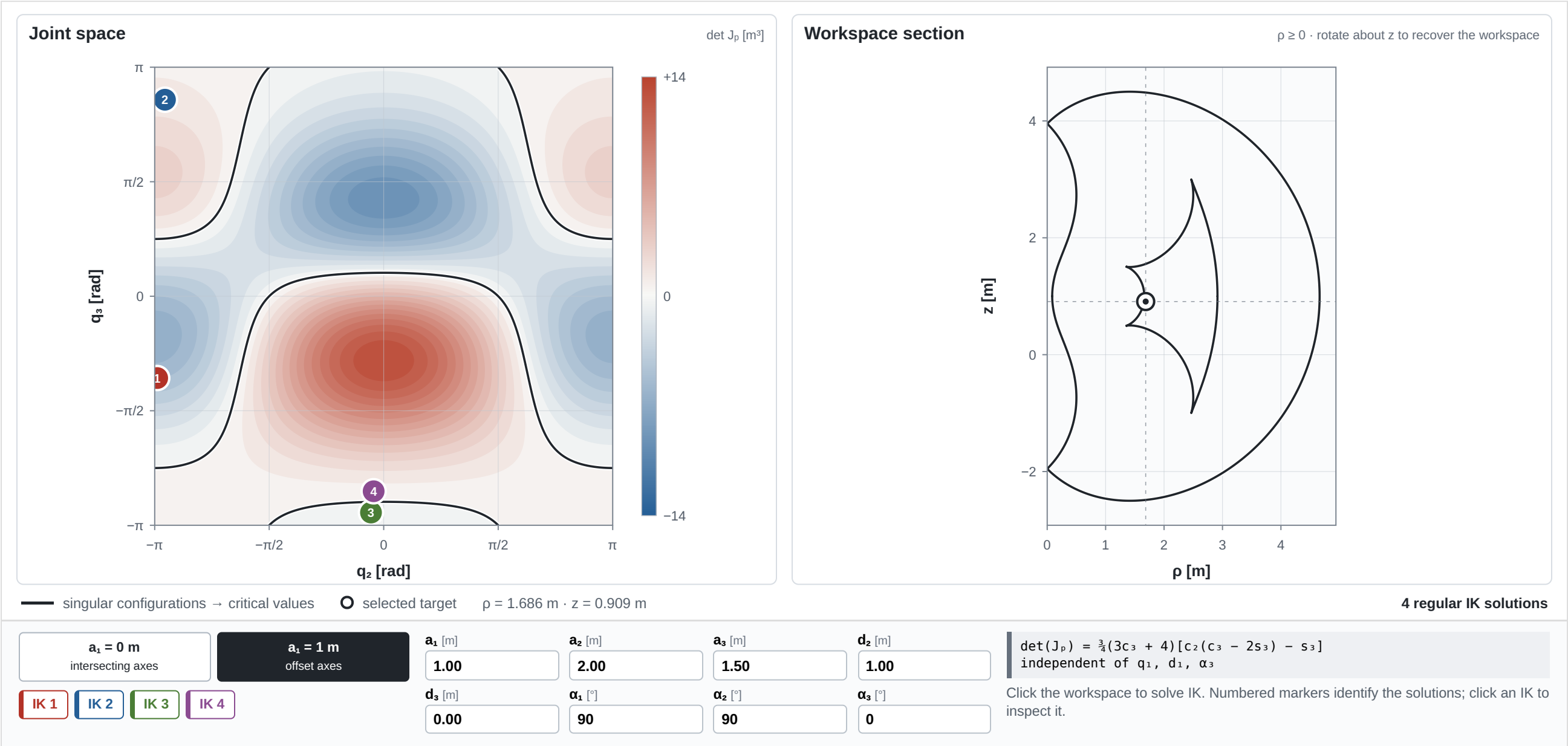


These viewers load the exact URDF kinematics and their visual meshes. The next atlas uses separate idealized D–H presets to isolate the effect of a_1 ; its metric geometry must not be inferred from these renderings.

Use idealized D–H presets to connect singular curves and IK counts

The left plot shows $D(q_2, q_3) = 0$; the right maps those configurations to critical values in (ρ, z) . These presets are analytical teaching models, not the exact URDFs shown previously.

Explore: switch the zero-offset and offset D–H presets, select workspace regions between critical curves, and compare their regular IK counts.



Cusps change how solution sheets connect

Regular value
Each plotted IK point is locally isolated and $D \neq 0$.

Fold
Two IK solutions merge on a regular critical curve and disappear across it.

Cusp
Three IK solutions coalesce. Encircling its critical value can permute solutions without meeting a singularity.

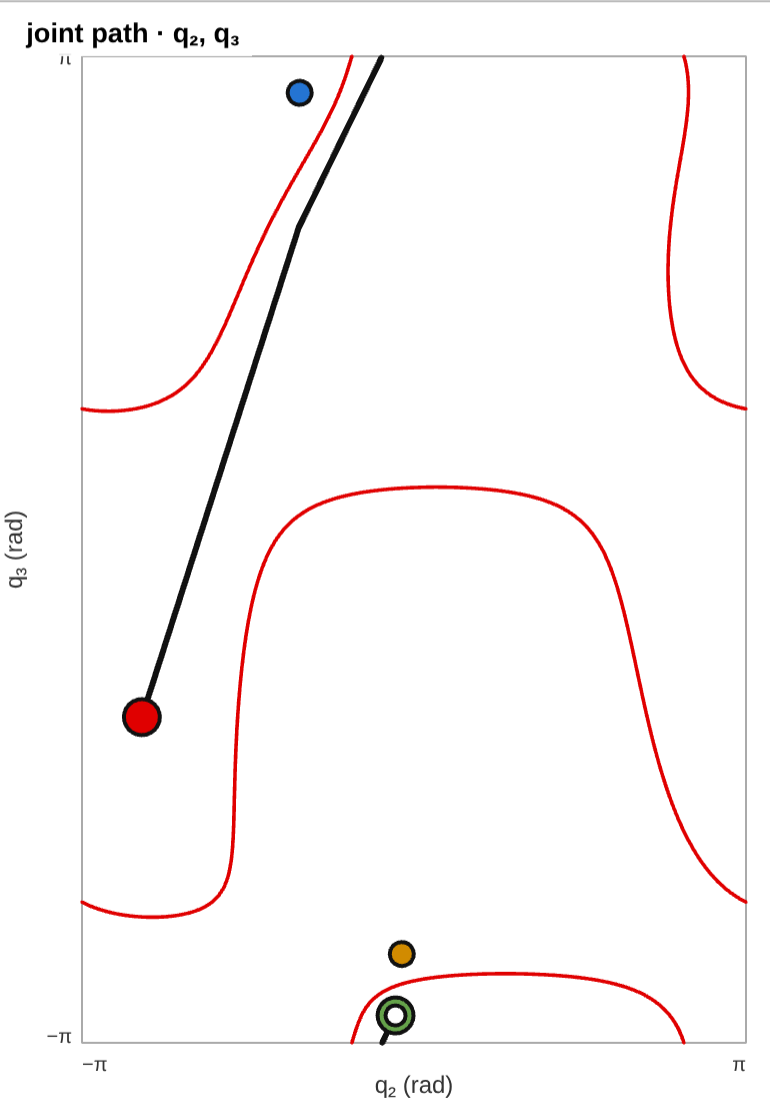
A cusp is not merely a sharp-looking workspace curve. It is a triple-root event in the inverse problem.

Build a joint path and watch its closed workspace loop

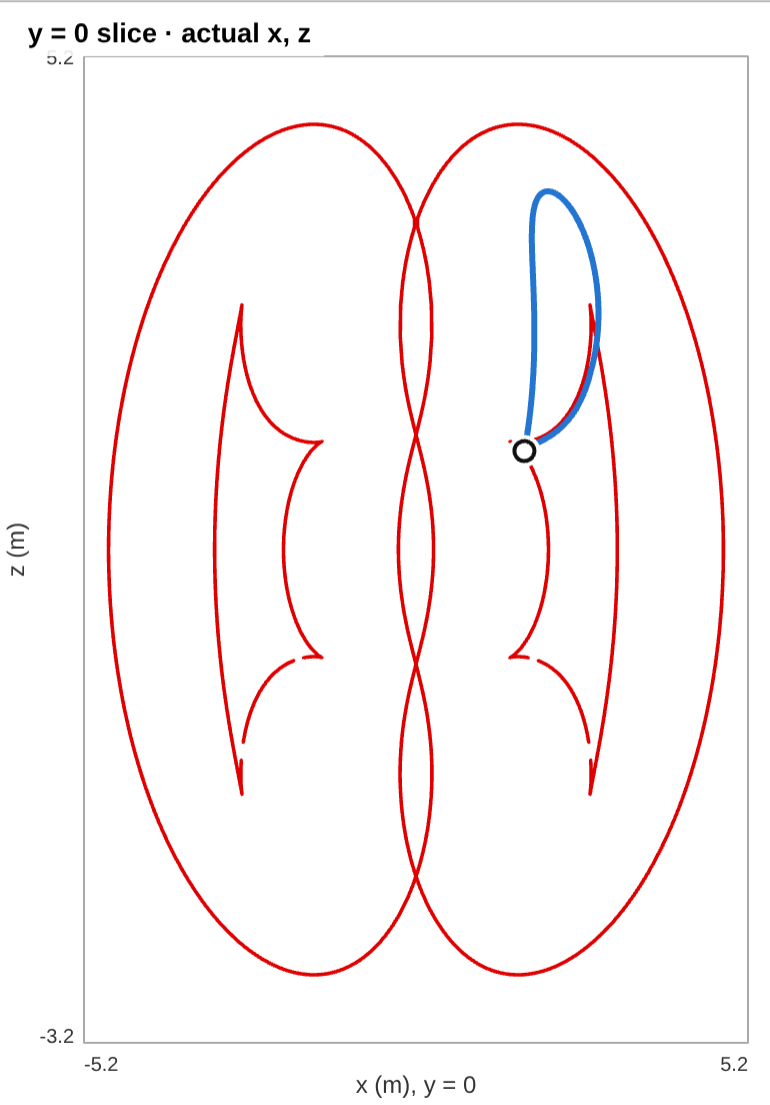
This laboratory uses the exact custom_3R_new.urdf kinematics and its visual meshes. A faded robot marks the starting configuration.

Explore: select $(x, 0, z)$ and two IKs. Build path searches the periodic joint grid, rejects singular edges, and samples joint-space interpolation. Play steps through these configurations in the $y = 0$ slice.

joint path · q_2, q_3



y = 0 slice · actual x, z

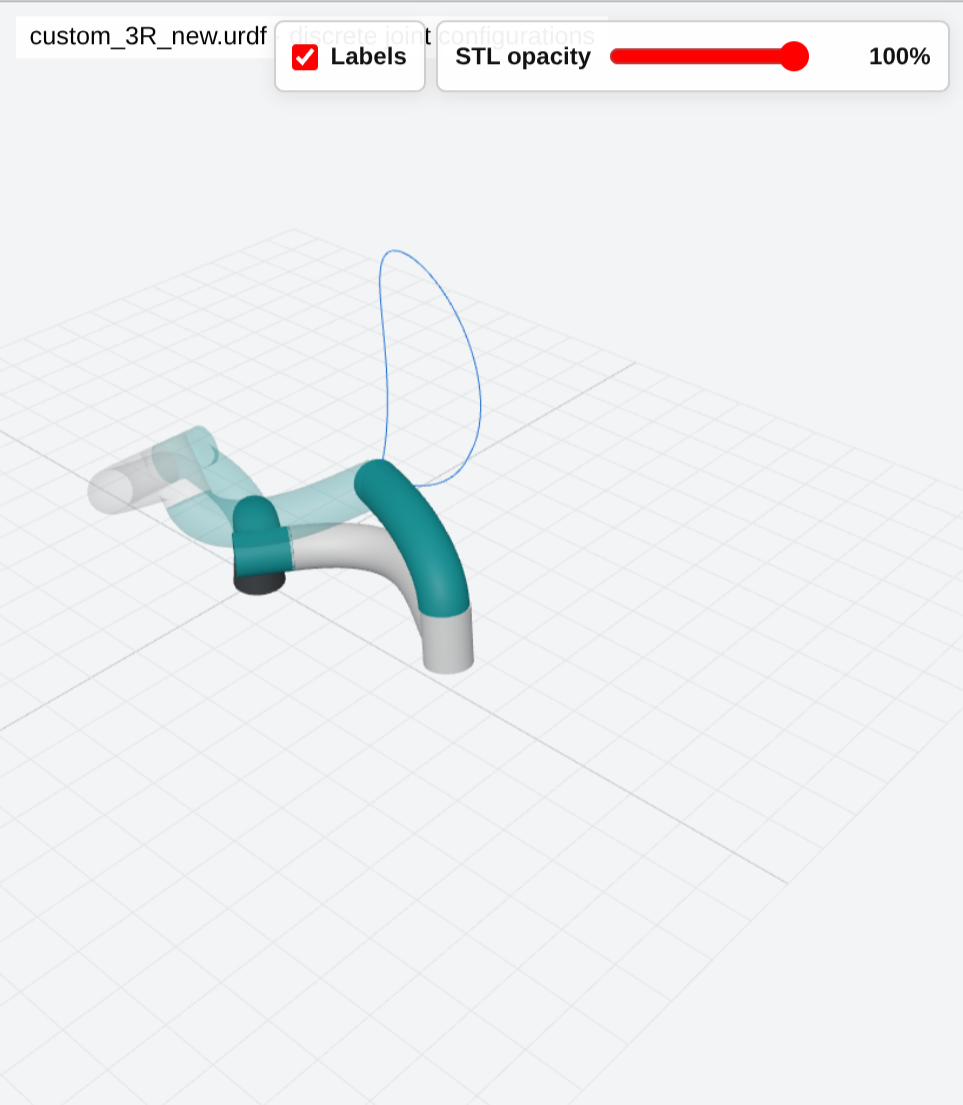


custom_3R_new.urdf

discrete joint configurations

☒ Labels

STL opacity 100%



IK 1

IK 2

IK 3

IK 4

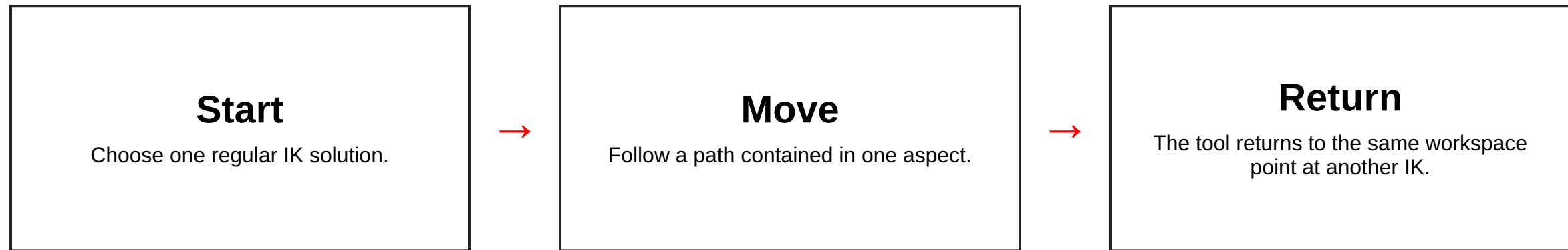
A* + checked shortcuts → 420 discrete joint samples. min |det J| = 0.374; endpoint error 3.7e-9 m. q₁ keeps each sample on y = 0. Press Play.

Build path

Replay

Reset to start

A validated nonsingular path demonstrates a change of IK solution



$$\det J_p(\gamma(s)) \neq 0 \quad \forall s \in [0, 1], \quad f(\gamma(0)) = f(\gamma(1)).$$

The ghost and final robot coincide at the tool, not in joint configuration. The sampled animation supports a nonsingular change of solution; a formal guarantee also requires a continuous lower bound away from rank loss along the whole path.

A cusp is a triple root of the inverse-kinematic polynomial

- 1** Use $t = \tan(q_3/2)$ to eliminate trigonometric terms and obtain $P(t; \rho, z) = 0$. Treat $q_3 = \pi$, represented by $t = \infty$, in a second chart.
- 2** Locate repeated-root candidates with $P = 0$ and $P' = 0$; their (ρ, z) projections are candidate critical values.
- 3** Impose $P'' = 0$. A root of multiplicity at least three is a cusp candidate.
- 4** Reject higher degeneracy, nonreal configurations, and roots introduced by cleared denominators: check $P''' \neq 0$ and substitute every candidate into the original kinematics.

$$P(t) = 0, \quad \frac{\partial P}{\partial t} = 0, \quad \frac{\partial^2 P}{\partial t^2} = 0$$

At a regular point of the singular curve, its tangent collapses under the reduced workspace map

$$g(q_2, q_3) = \begin{bmatrix} \rho \\ z \end{bmatrix}, \quad D = \det J_p, \quad \boldsymbol{\tau} = \begin{bmatrix} D_{q_3} \\ -D_{q_2} \end{bmatrix}$$

$D = 0, \quad \nabla D \neq \mathbf{0}, \quad J_g \boldsymbol{\tau} = \mathbf{0}$

This is a test in the chosen (q_2, q_3) reduction, not a coordinate-free formula.
Confirm every numerical candidate with the triple-root conditions.

Offset-atlas result

For $a = (1, 2, 1.5)$, $d = (1, 1, 0)$, and $\alpha = (\pi/2, \pi/2, 0)$:

q_2	q_3	ρ	z
-129.783°	66.385°	2.4656	-0.9987
-78.596°	-171.935°	1.3555	0.4953
78.596°	-171.935°	1.3555	1.5047
129.783°	66.385°	2.4656	2.9987

A spherical wrist separates position and orientation again

$${}^3J_5 = \begin{bmatrix} J_{A,v} & 0 \\ J_{A,\omega} & J_{W,\omega} \end{bmatrix}, \quad \det({}^3J_5) = \det J_{A,v} \det J_{W,\omega}.$$

Positional subchain

Joints 1 to 3 place the wrist point O_5 . Their singularity set can carry cusp points.

Wrist module

Joints 4 to 6 orient the tool about the wrist center. Its two regular flip solutions are separated by $\sin q_5 = 0$.

The spherical-wrist module is noncuspidal

Two wrist solutions

$$(q_4, q_5, q_6) \quad \text{and} \quad (q_4 + \pi, -q_5, q_6 + \pi)$$

The separator

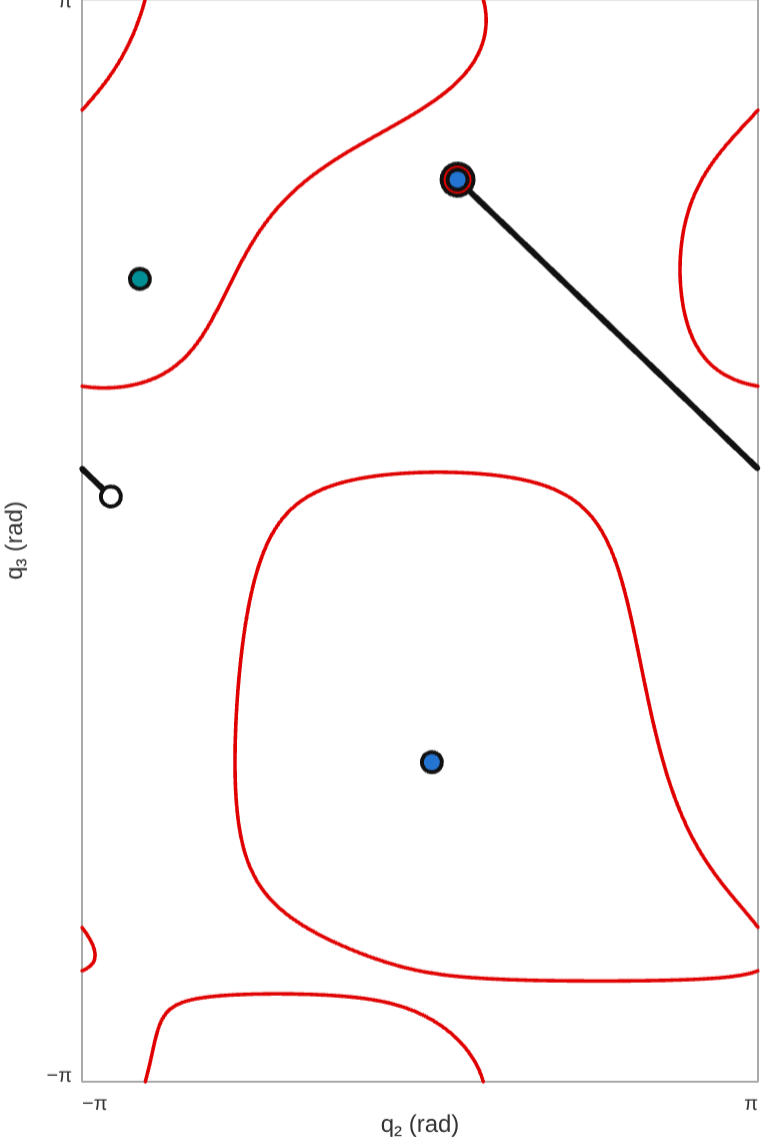
Any continuous change between the flips changes the sign of $\sin q_5$. It must therefore cross $q_5 = 0$ or $q_5 = \pi$, where axes 4 and 6 align.

This conclusion applies to the wrist module. For an ideal wrist-partitioned 6R chain, whole-robot cuspidality is then governed by the positional 3R subchain, subject to the assumed decoupling and unconstrained configuration space.

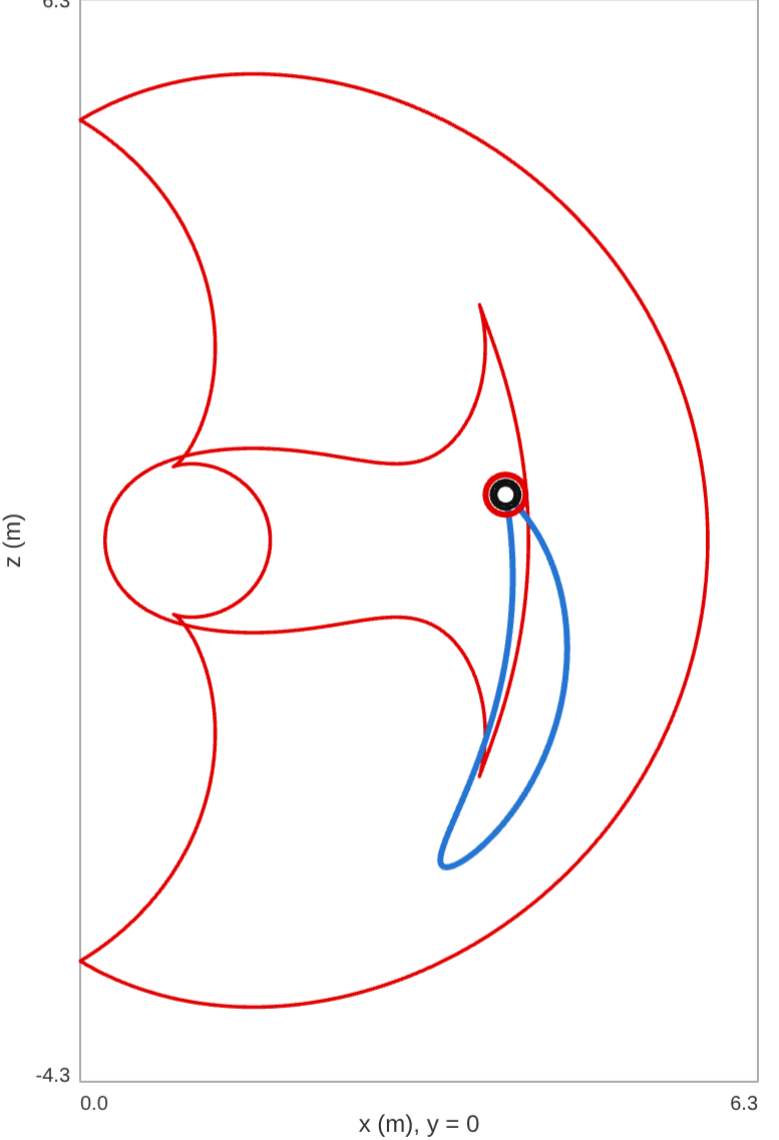
custom_6R_new.urdf inherits its cuspidality from joints 1 to 3

Select the wrist center $O_w = (x, 0, z)$, then choose start and goal IKs. Build path searches the positional chain and continuously adjusts the wrist to preserve the same tool orientation.

positional singularities · q_2, q_3



$y = 0$ · wrist-center workspace slice



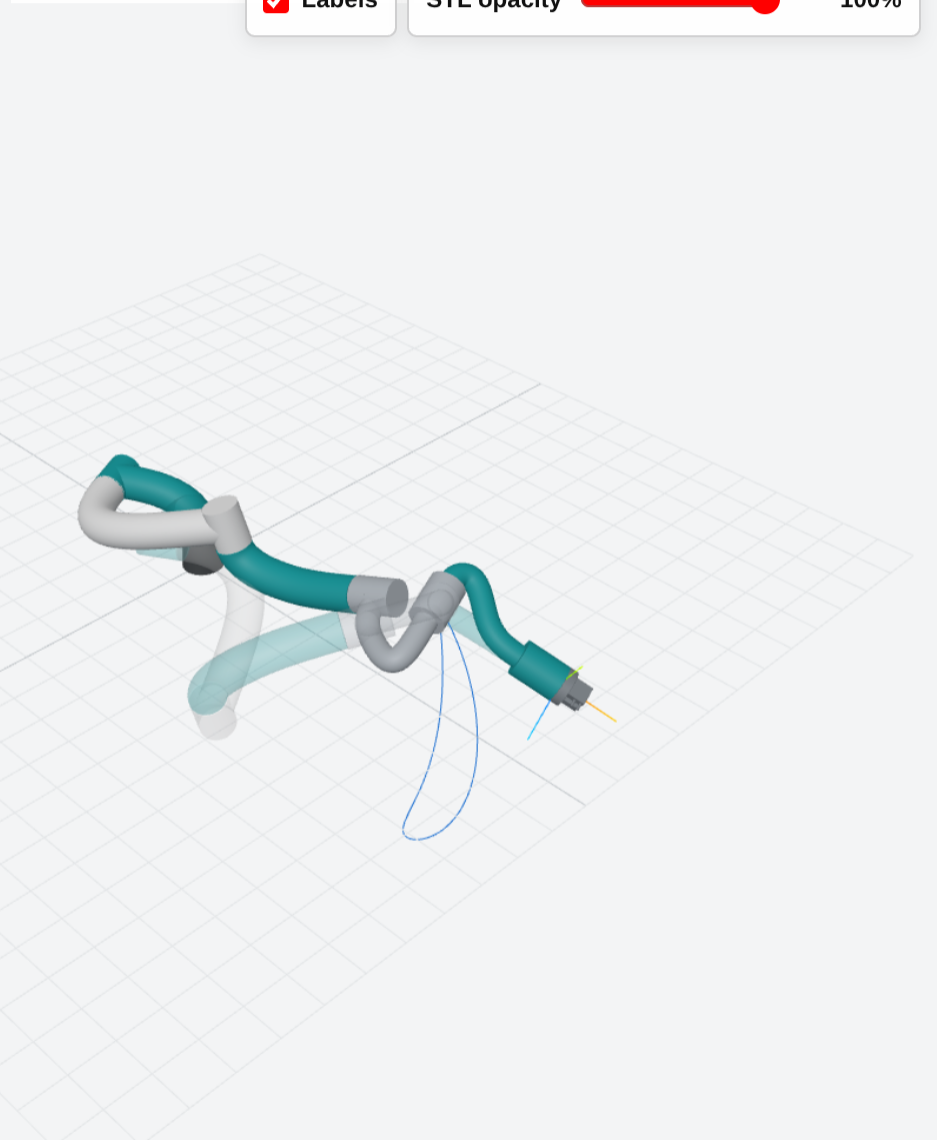
fixed tool orientation

BGR tool frame

☒ Labels

STL opacity

100%



IK 1

IK 2

IK 3

IK 4

IK 5

IK 6

IK 7

IK 8

448 discrete configurations · min |det J| 0.656 · min |sin q_5 | 0.152 · max rotation error 1.8e-15. Arm edges and the continued wrist pass numerical checks. Press Play.

Start

IK 1

Goal

IK 3

Build path

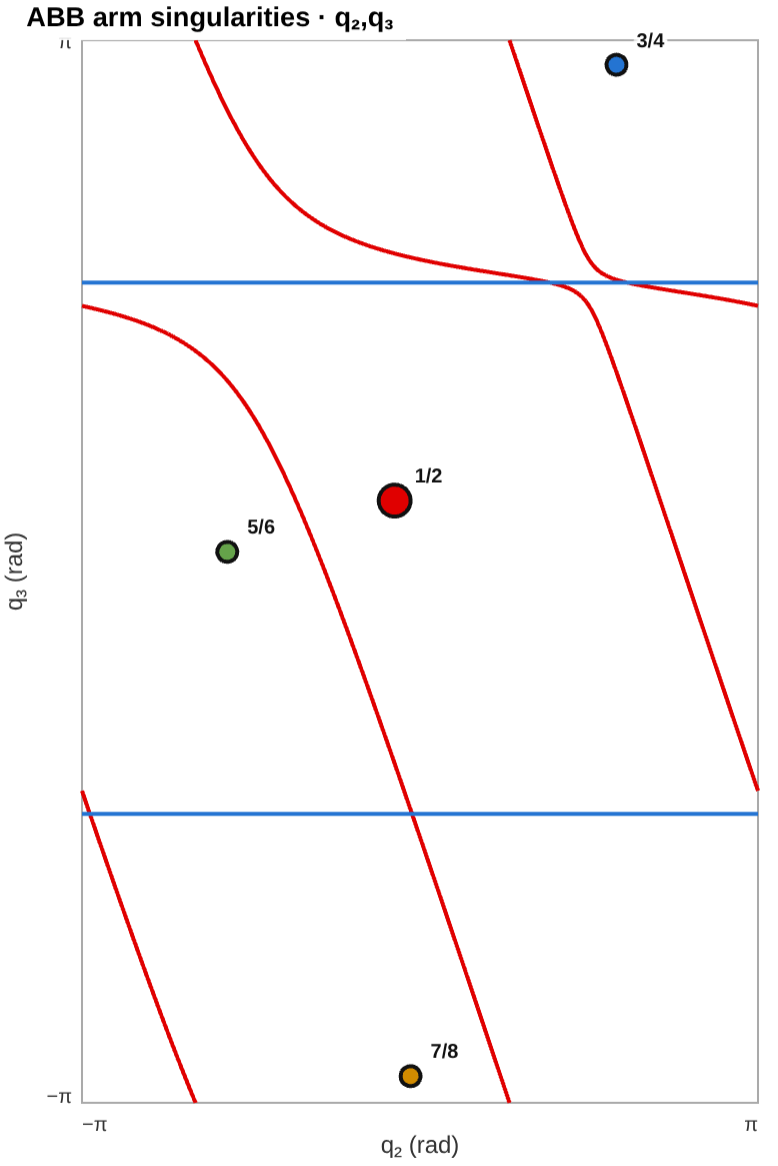
Replay

Path

Explore the ABB IRB 4600 singularities and IK distribution

Keep the tool z-axis vertically downward. Select a wrist center ${}^0\mathbf{p}_5 = (x, 0, z)$, then compare its shoulder, elbow and wrist branches. The example target is used again in the following tables.

ABB arm singularities · q_2, q_3



— shoulder: $G = 0$

— elbow: $F = a_3 \sin q_3 + d_4 \cos q_3 = 0$

wrist: $\sin q_5 = 0$ · coincident arm markers represent wrist flips

wrist-center section · $y = 0, x \geq 0$

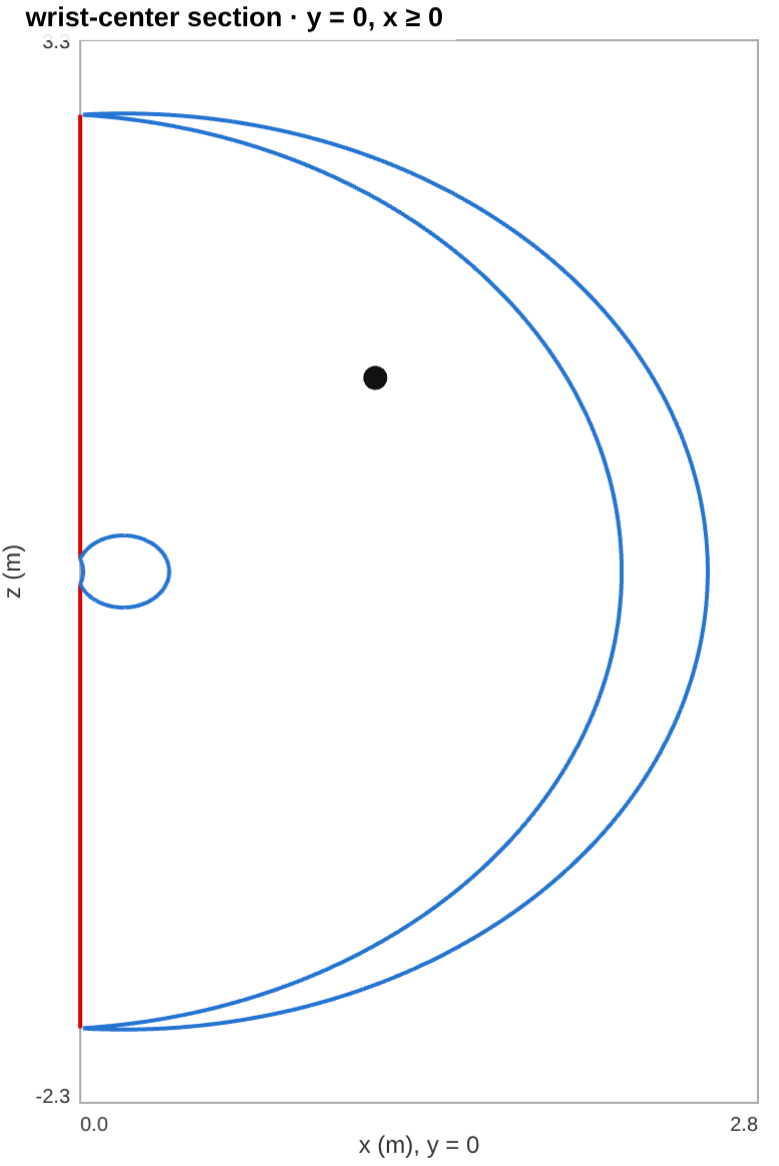
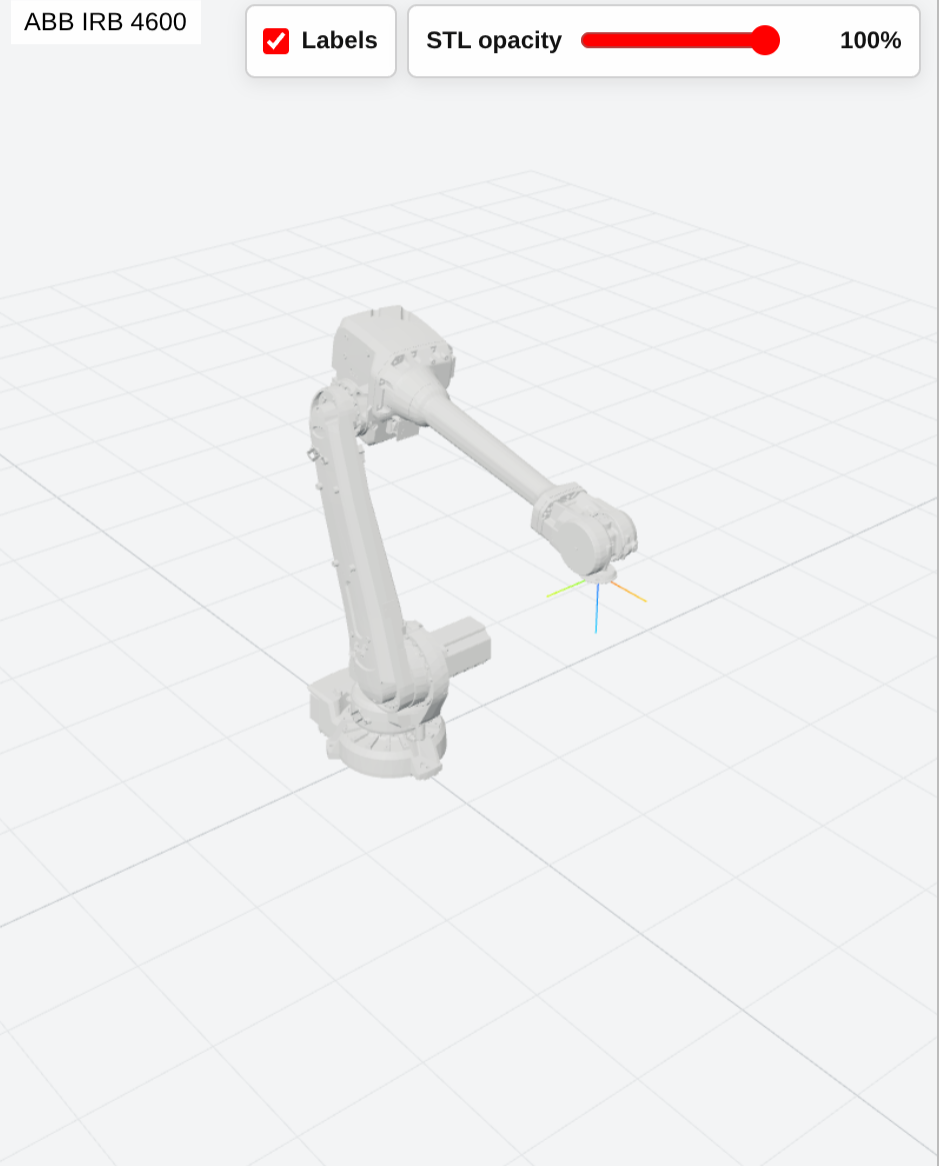


ABB IRB 4600

☒ Labels

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IK 1

IK 2

IK 3

IK 4

IK 5

IK 6

IK 7

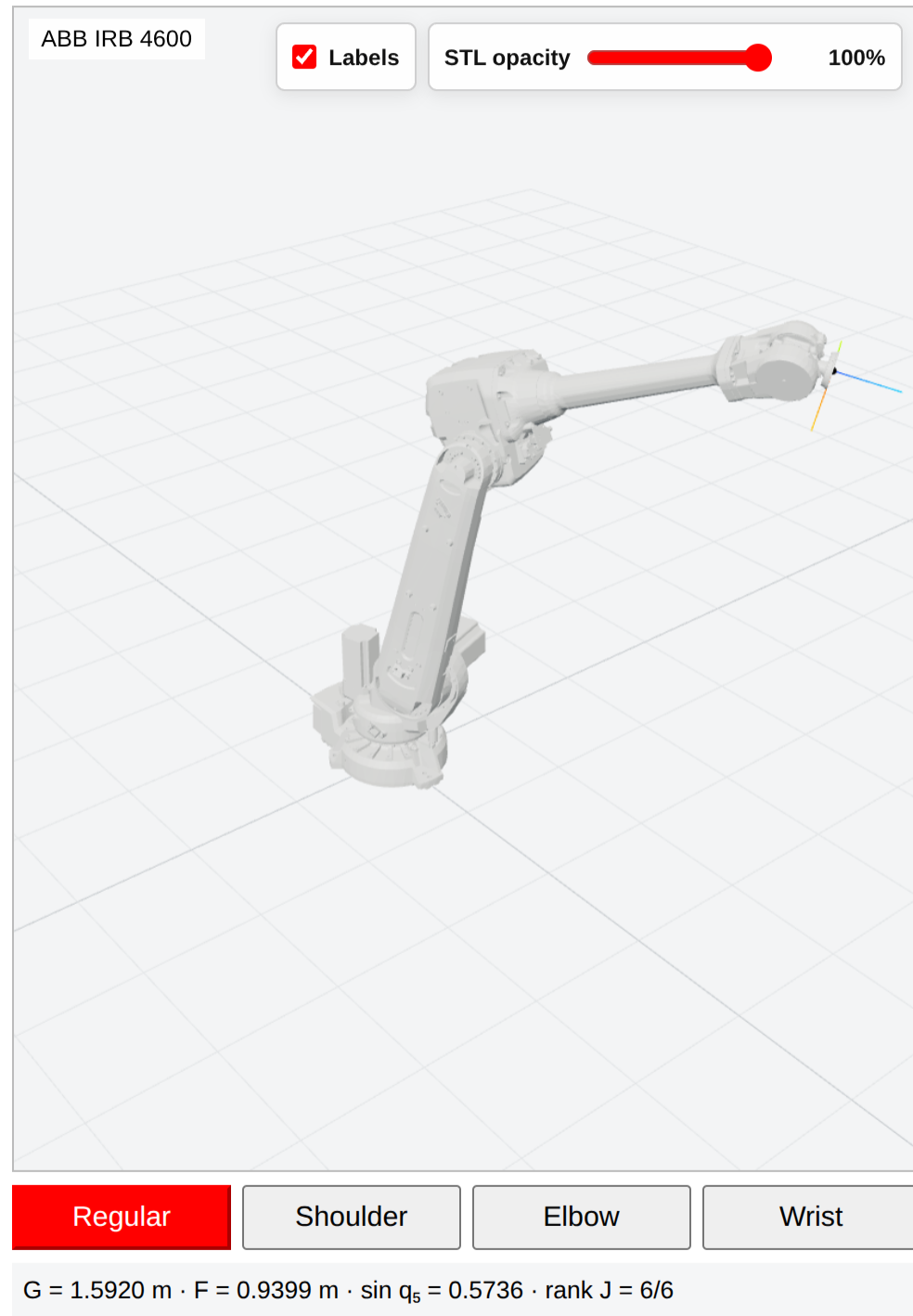
IK 8

8 regular IKs · IK 1: S +, E +, W + · wrist error 1.6e-17 m · rotation error 1.2e-16. Mathematical branches before joint-limit filtering.

Example target

☐ cycle through IKs

The ABB IRB 4600 has three binary singularity separators



$$\det({}^0J_5) = -a_2 G F \sin q_5$$

shoulder · G

elbow · F

wrist · $\sin q_5$

$$G = a_1 + a_2 s_2 + a_3 s_{23} + d_4 c_{23},$$

$$F = a_3 s_3 + d_4 c_3.$$

Here $s_{23} = \sin(q_2 + q_3)$ and $c_{23} = \cos(q_2 + q_3)$. As in Lecture 05, $a_1 = a_3 = 0.175$, $a_2 = 1.095$, $d_4 = 1.270$, and $d_1 = 0.495$ m.

- **G = 0**: the wrist center lies on the base axis.
- **F = 0**: $q_3 \approx -82.154^\circ + k 180^\circ$; the arm loses one translational direction.
- **$\sin q_5 = 0$** : axes 4 and 6 align.

Each factor's sign stays constant along a nonsingular path. Changing a shoulder, elbow or wrist sign requires crossing its zero set.

One ABB IRB 4600 target has eight regular IK solutions

Wrist center ${}^0\mathbf{p}_5 = (1.200, 0, 1.500)$ m; ${}^0R_t = \text{diag}(1, -1, -1)$. With $d_6 = 0.135$ m, the tool origin is ${}^0\mathbf{p}_t = (1.200, 0, 1.365)$ m.

IK	q_1	q_2	q_3	q_4	q_5	q_6
1	0.000	-13.499	24.016	0.000	79.483	180.000
2	0.000	-13.499	24.016	-180.000	-79.483	0.000
3	0.000	104.628	171.675	0.000	173.697	180.000
4	0.000	104.628	171.675	-180.000	-173.697	0.000
5	180.000	-102.650	6.660	-180.000	174.010	180.000
6	180.000	-102.650	6.660	0.000	-174.010	0.000
7	180.000	-5.023	-170.969	-180.000	94.008	180.000
8	180.000	-5.023	-170.969	0.000	-94.008	0.000

Angles are in degrees, in the same convention as Lecture 05. These are eight mathematical branches of one pose; rows 1, 2, 7 and 8 satisfy the robot's joint limits. Values shown are rounded.

Each ABB IK receives its shoulder, elbow and wrist signs

For the preceding target: $S = \operatorname{sgn} G$, $E = \operatorname{sgn} F$, and $W = \operatorname{sgn}(\sin q_5)$.

IK	S = sign G	E = sign F	W = sign sin q ₅
1	+	+	+
2	+	+	−
3	+	−	+
4	+	−	−
5	−	+	+
6	−	+	−
7	−	−	+
8	−	−	−

The eight IKs have eight different sign triples. Any path between two of them must change at least one sign and cross a singularity. This ABB geometry is noncuspidal.

The classification relies on both branch count and factorization

Eight solutions

Three binary decisions can name all regular branches.

Three factors

Each determinant factor supplies a genuine singular separator associated with one label.

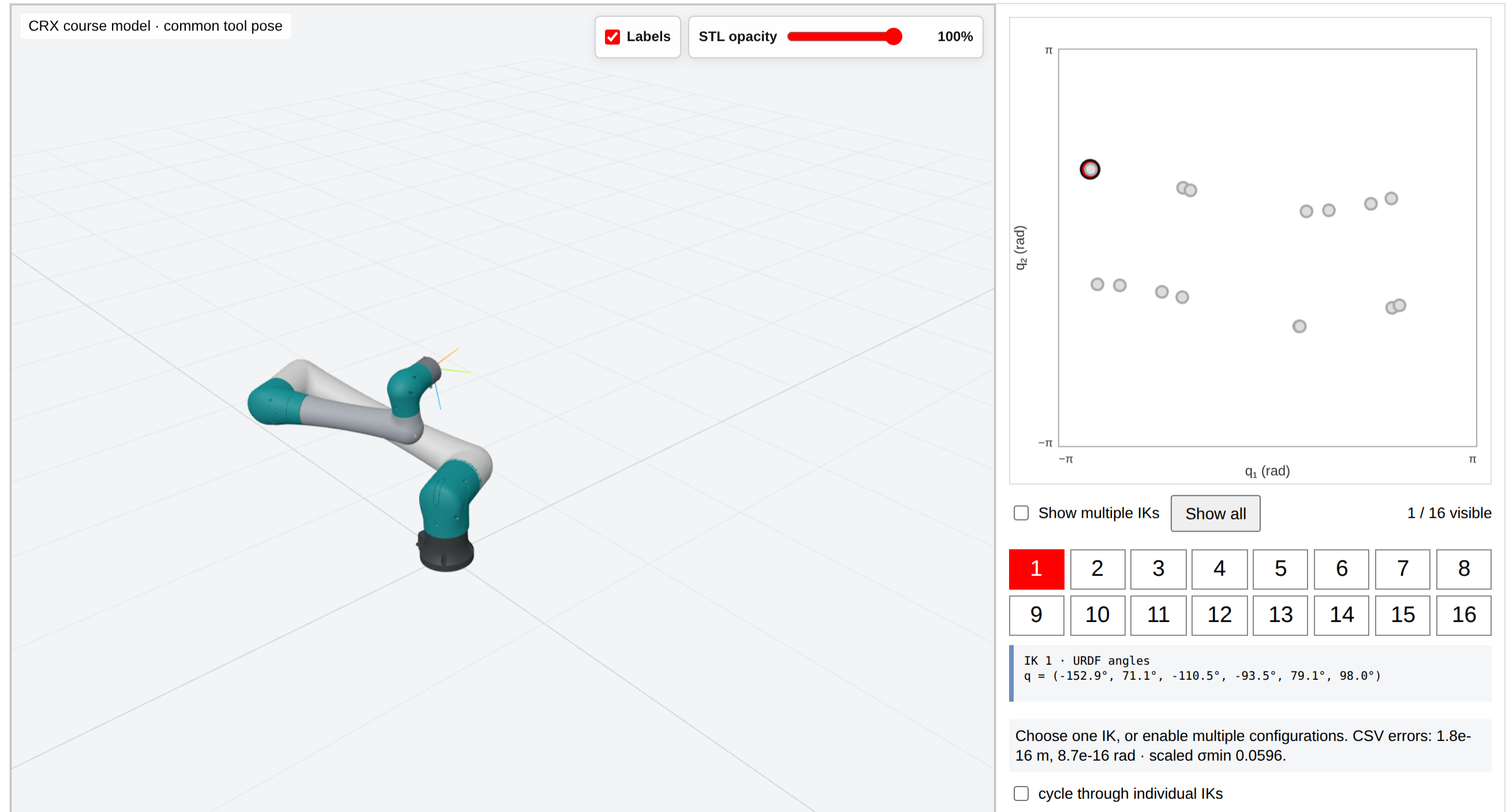
Beyond this structure

More solutions or coupled factors require topology, continuation, and path analysis rather than a posture mnemonic.

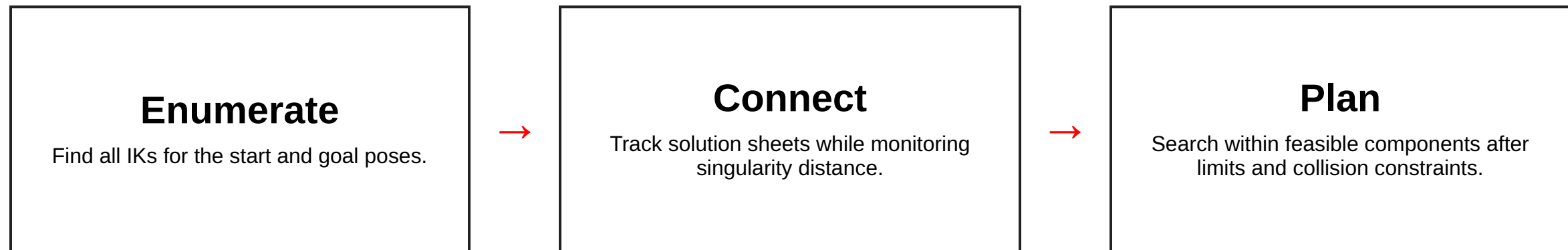
One FANUC target provides sixteen regular IK candidates

The supplied solution set is mapped from its standard D–H angle convention into the course CRX model.

Matching one fixed target demonstrates the solution count; it does not prove how those solutions connect.



A branch label is not a certificate of connectivity



Cuspidality is a global property of IK connectivity

Distinct IKs of the same pose can belong to one aspect. Understanding their connections requires following **whole paths** through joint space.

Feasible path

The chosen starting IK can follow the prescribed tool path continuously, under the imposed constraints.

Infeasible path

That starting IK cannot complete the prescribed path. Another starting IK may allow it.

Nonsingular change of solution

A tool loop ends at a different IK without a singularity. Such a loop may be repeatable, cycling through IKs.

Nonrepeatable path

A loop works once, but its continuation cannot be repeated indefinitely from the configurations it reaches.

Why planning is urgent: many modern collaborative robots are cuspidal. Their global IK connectivity calls for path-planning algorithms that handle these different possibilities.